BORDER SITE · ON-PREMISE AT EACH OF THE 7 UAE ENTRY POINTS · RAW IMAGES NEVER LEAVE

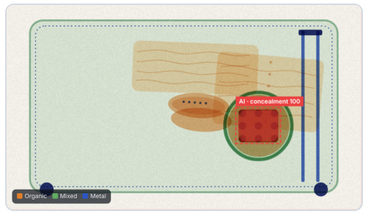
X-ray scanner
Dual-energy belt / cargo scanner
EXISTING

Desk camera
Officer tablet at the inspection desk
EXISTING

Hyperspectral camera
SWIR, 1000–2500 nm, at the desk
ADD-ON

Edge AI box
Small GPU computer beside the scanner; runs M1–M3 in < 1 s per bag, works offline
NEW · SOFTWARE-DEFINED
model updates (B)

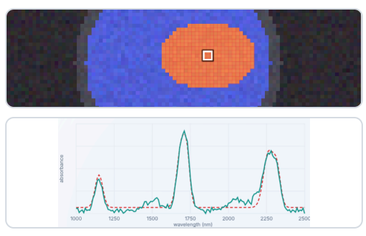
AI modules on the edge box



M1 X-ray Vision
Fully-convolutional segmentation net (28.7k params)
Pixel-level concealment heatmap + bag score
Test (600 bags): 100% recall, 1.7% false alarms



M2 Pills, Vials & Labels
Tablet-imprint CNN (88.1k params, 99.8% accuracy)
Arabic/English label OCR + registry check
Physical profile from the mm scale



M3 Hyperspectral ID
Identifies the substance through packaging
Pearson match to drug library + NNLS mixture unmixing
Library updated by M4 (A)

Officer in the loop
Sees the heatmap, evidence and fused score on the tablet; decides **clear · re-scan · open · seize**.
Every action is logged (audit trail); supervisor approves seizures.
Seized sample goes to the lab for legal confirmation → (1)

evidence

Manifest & booking data
Route-pattern prior · no nationality / ethnicity

INSPECTION CASE · ONE PER BAG / PARCEL
C-24133 · BAG-T3-118211

M5 Risk Fusion

Route prior		0.20
M1 X-ray		0.75
M2 Tablets		0.50
M3 Spectrum		0.55

$$S = 1 - \prod_i (1 - w_i\, p_i)$$
weighted noisy-OR evidence fusion

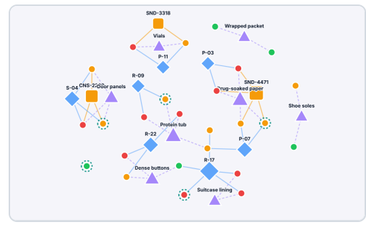
92
Fused risk score
Each module's share of the score is shown to the officer.

Clear | Re-scan | Open bag | Seize

Every module writes to the same case: a single, time-stamped evidence timeline that becomes the court-ready **evidence pack** (PDF with chain of custody).

Illustrative example values.

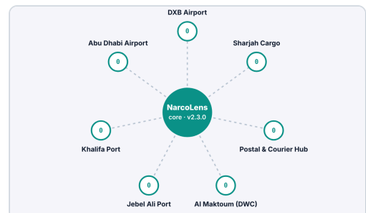
SOVEREIGN CORE · ON-PREM GOVERNMENT DATA CENTRE



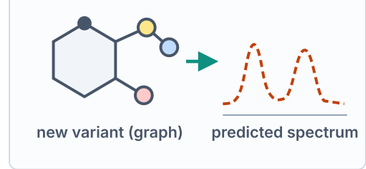
NET Network analytics
Links cases across sites by sender, route pattern and concealment method
Graph clustering → auto-generated insights for investigators



Command & supervisor apps
Live map and KPIs for all 7 entry points
Seizure approvals, override review, audit log
Shift heat calendar for K9 placement



FL Federated learning server (B)
Secure aggregation of site updates (Δ only)
Differential privacy (ε = 1–8); poisoned updates rejected
Signed models rolled out to every edge box



M4 Novel-Drug Predictor (A)
Graph neural network predicts the spectral fingerprint of new synthetic variants
Chemists validate → pushed to every M3 library *before* the first seizure

CONTINUOUS LEARNING LOOP · RUNS AFTER EVERY LAB-CONFIRMED SEIZURE

signed models → edge boxes B

1. **Lab confirms the seizure** — Forensic result = legal proof and the training label
2. **Local retraining on site** — Edge box fine-tunes M1–M3 on its own confirmed cases
3. **Clip · add noise (ε) · encrypt Δ** — Only the model update leaves the site, never images
4. **Robust secure aggregation** — Core averages updates; outliers / poisoning rejected
5. **Signed model to all 7 sites** — Every site learns from every seizure

at the border site — in the sovereign core

→ data / evidence (only case metadata reaches the core) ⇢ model or library updates (A) (B) connector to the matching module EXISTING hardware customs already owns M1 NarcoLens AI module

Figure 1. NarcoLens system architecture. Each UAE entry point runs an edge AI box beside its existing scanners. M1–M3 add evidence to a single *Inspection Case* per bag or parcel, and M5 fuses it into one explainable risk score; the officer makes the decision. Only case metadata and encrypted, noised model updates leave a site. The sovereign core links cases across sites, predicts new synthetic-drug variants (M4) and aggregates federated updates so that every site learns from every seizure. In the prototype, M1 and M2 are trained neural networks, M3 and M5 are live algorithms, and M4 and federated learning are simulated. All images are outputs of the working prototype on synthetic data.